Glacier-Wide Mass Balance and Compiled Data Inputs

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Overview: This document describes the data associated with the long-term mass balance measurement campaigns at the USGS Benchmark Glaciers, including Gulkana, Wolverine, Lemon Creek, South Cascade and Sperry. Data collected on other glaciers that conform to the field data collection and data processing workflow of the USGS Benchmark Glacier Project standards are included where available.

SUMMARY

This data release contains glacier mass balance data collected on various North American glaciers under various projects and funding sources (Figure 1). The following summary describes the full suite of data included for USGS Benchmark Glaciers. Data on other glaciers of interest conform to Benchmark Glacier Project data processing standards, but may not comprise the full input-output data suite described herein. Since the late 1950s, the USGS has maintained long-term glacier mass-balance projects at several North American glaciers. Measurements began on South Cascade Glacier, Washington in 1958, expanding to Gulkana and Wolverine glaciers, Alaska in 1966, and later Sperry Glacier, Montana in 2005. The Juneau Icefield Research Program has measured glacier mass balance on Lemon Creek since the mid-1940s, and USGS initiated seasonal measurements at Lemon Creek in 2014 (McNeil and others, 2020). Measurements and data on other glaciers are included where available. Seasonal and annual glacier-wide mass balances are estimated where direct field measurements of point glaciological data, weather, and geodetic data are available. The analysis framework (O'Neel and others, 2019; Florentine and others, 2023; prior to v 3.0 van Beusekom and others, 2010) is identical at each glacier to enable cross-comparison between output time series. Vocabulary used is defined herein and follows Cogley and others (2011) *Glossary of Glacier Mass Balance and Related Terms*.

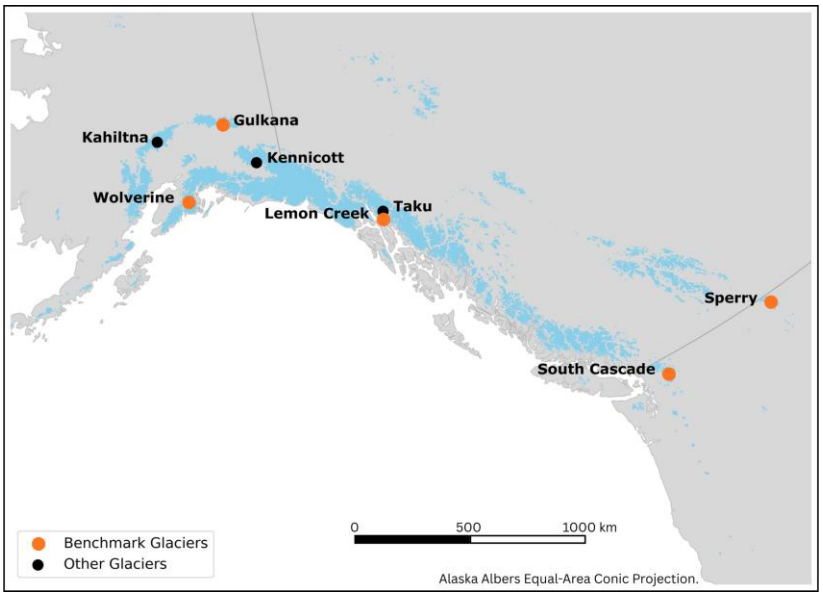


Figure 1) Location of glacier mass balance data contained with this data release. Green dots indicate the USGS Benchmark Glaciers (Gulkana, Wolverine, Lemon Creek, Sperry, and South Cascade glaciers). Black dots represent other glaciers of interest, and blue areas represent all glaciers in the region (RGI consortium, 2017)

This data release includes compiled input data and glacier wide mass balances. Input data are of three types: 1) time-variable area altitude distribution (AAD); 2) time series of point mass balance; 3) weather data from glacier-adjacent USGS operated weather stations and nearby long term weather stations operated by others (FAA, USDA) and retrieved from NOAA National Centers for Environmental Information. Output data represent conventional glacier-wide estimates of surface mass balance which are bias corrected with geodetic calibration. Mass balances are calculated as meaning mass divided by area and are reported in water equivalent depth (e.g., meters of water of equivalent, m w.e.). Whenever possible, site mass balances average multiple field measurements to account for

local surface roughness and limit the influence of measurement error. Additional data are available in the larger USGS Glacier Project Comprehensive Data Collection, available at https://doi.org/10.5066/P9AGXQSR.

The most recent year of mass balance estimates are preliminary as any ablation after the date of the fall visit is modeled rather than measured. During subsequent field visits, any ablation that occurred over the winter season (winter ablation) is measured and used to revise the previously modeled estimate of mass balance.

PURPOSE

The purpose of this project is to quantitatively evaluate changes in mass over time at specified glaciers.

PROJECTION AND DATUM:

Individual glacier site locations for glaciological and weather data used within this data release, are provided within *Glacier_Mass_Balance_Sites.zip* in shapefile (.shp), Keyhole Markup Language (.kml), and comma separated value (.csv) formats. Each site's Universal Transverse Mercator (UTM) 'Easting' and 'Northing' coordinates are provided in the respective UTM zone EPSG code listed in 'UTM_EPSG'. Geographic Coordinate System (GCS) World Geodetic System (1984) latitude and longitude are also provided using EPSG code 4326.Elevations are referenced to the WGS84 ellipsoid.

UNITS:

The "meter water equivalent" (m w.e.), describes *glacier* mass in *specific* (per unit area) units as the thickness of an equal mass having the *density* of water. The meter water equivalent is obtained by dividing a particular mass per unit area by the density of water: 1 m w.e. = 1000 kg m^{-2} / ρ_W (Cogley and others, 2011).

SITES

Site locations are given in the file *Glacier_Mass_Balance_Sites.zip* as ESRI Shapefile (.shp), Keyhole Markup Language (.kml), and Comma Separated Value (.csv) formats. Within these files, column "Glacier" indicates the glacier's name of each site; "Type" reflects if the site is a source of glaciological or weather data; "Site" denotes a unique identifier of each site on a given glacier; "Easting" and "Northing" Coordinates are given for the appropriate UTM zone appropriate for each location, and listed under 'UTM_EPSG"; "Latitude" and "Longitude" are also provided for each site using GCS WGS 1984 EPSG 4326, and listed under "LatLon_EPSG". UTM zones range from UTM Zone 5N to 12N. Some stake locations have been less consistent over the period of record, becoming more consistent in the last 10-20 years.

DATA

Input and output data files are presented for each glacier in CSV format. Within these files, 'nan' indicates the lack of a measurement or an unresolved value while a value of 0 means a measured zero. Data files and file headers are described below. File naming convention is presented where *Glacier* is replaced with the glacier name (e.g. Sperry or Gulkana) in file names. In the descriptions below, data file names are enumerated while column headers of those files are described in the bullets.

INPUT DATA

I Input data files required for glacier wide mass balance calculations are described in sections **1**-**4**. Additional data that further constrain the mass balance data processing routine are provided when available and described in sections **5** and **6**.

1. **Input_*Glacier*_Glaciological_Data.csv**
   File containing time series of point water balance measurements with the following columns:
   - **Year:** Calendar year in which the annual mass balance ends.

- **site_name:** Describes glacier mass balance measurement site names. USGS_Glacier_Sites.zip contains the coordinates for these sites.
- **spring_date**: Date of spring measurement.
- **fall_date:** Date of fall measurement.
- **elevation:** Elevation (m) of the measurement site's height above the WGS84 ellipsoid.**bw_snow_depth:** Depth of seasonal snow accumulation in meters, measured on the spring date, and used to calculate the bw value.
- **bw_density:** Snow density in kg/m^3 of the seasonal snow accumulation at the site. This column contains 'nan' values where the density was not measured at the site.

- **bw**: Winter mass balance at the site estimated from stake, pit, core, and/or probe measurements provided in units of m w.e. Measurements are stratigraphic in the sense that they are made from the surface at the time of measurement down to the previous summer's surface. Values provided in this file may use an average of multiple measurements made near the site to represent the spatial variability.
- **ba_snow_depth:** Snow depth in meters of any remaining annual accumulation, measured on the fall date, and used to calculate the value of ba. This column is left blank where the value of ba is negative.
- **ba_density:** Snow density expressed in kg/m^3 of any remaining annual accumulation measured on the fall date and used to calculate the value of ba. This column contains 'nan' values where the density was not measured at the site.
- **ba**: Annual (net) mass balance at the site estimated from stake and/or pit measurements provided in units of m w.e. Measurements are stratigraphic in the sense that they are made from the current surface (or, if there is new snow, they are made from this summer's now buried surface) down to the previous summer's surface. Summer mass balances may be derived as the difference between annual and winter mass balance. Values provided in this file may use an average of multiple measurements made near the site to represent the spatial variability.
- **winter_ablation**: Ablation (m w. e.) that occurred after the fall field visit and was measured the subsequent spring visit. Although measured in balance year i this ablation occurred in balance year *i*-1. We partition the balance in this way to be clear about the timing of the measurements. To convert to a measurement-interval mass balance (i.e., the net mass balance from the date of the previous fall measurement to the date of the spring measurement. See Cogley and others, 2009), winter ablation should be subtracted from the bw and the ba. In the case where winter ablation was not measured on the spring field visit in year *i*, and new snow was not present on top of the summer surface in the fall field visit of balance year *i*-1, winter ablation is represented as 'nan' given an unmeasured amount of additional ablation after the fall measurement date.
- **summer_accumulation_snow_depth:** New snow (meaning snow that has accumulated on the surface above this summer's melt surface) depth in meters measured on the fall date and used to calculate summer accumulation values. This column is left blank where no new snow is on the surface
- **summer_accumulation_density:** New snow (snow that has accumulated on the surface since this summer's melt surface) density in kg/m^3 of any new snow at the site. This column contains 'nan' values where the density was not measured at the site.
- **summer_accumulation:** Accumulation (m w.e.) that occurred before the fall field visit but was measured at the time of the fall visit. Although measured in balance year *i* this accumulation applies to balance year *i*+ 1. We partition the balance in this way to be clear about the timing of the measurements. To convert to a measurement interval mass balance (i.e., the net mass balance from the date of the previous fall measurement to the date of the spring measurement. See Cogley and others, 2009), summer accumulation should be added to the current year ba and subtracted from the following year's bw and ba. If new snow existed but was not measured, summer accumulation is represented as 'nan'.

Uncertainties: For all input values, uncertainties are poorly constrained, and evaluated primarily on the basis of expert opinion. Uncertainty sources and magnitudes are cited accordingly.

- *Snow depth:* The uncertainty of individual snow-depth measurements is generally small and is primarily controlled by instrument precision and local surface roughness over ice. Over firn, additional uncertainty may arise from misidentification of the previous summer surface; this represents the potential for mismeasurement rather than random measurement uncertainty. The dominant source of uncertainty in site-averaged snow depth arises from spatial variability within the measurement site and the representativeness of the sampling strategy, rather than the precision of individual measurements. Multiple snow-depth measurements using a combination of techniques (e.g., snow pits or cores, stakes, and probe measurements) are therefore collected within a site to better sample this variability and improve confidence in the representativeness of the resulting site-average value. While uncertainty estimates for individual snow-depth measurements are not commonly reported, previous studies indicate that uncertainty associated with site-scale representativeness (due to spatial variability or misidentification) is on the order of ~0.5 m (e.g., Beedle and others, 2009; Huss and others, 2009; Thibert and others, 2008).
- *Snow density:* Measurement uncertainty of density is related to sampling methods but is similar for both snow-pits and cores. In snow-pits, 1000 mL RIP cutters that we use have been shown to have a bias of up to 6%, with the bias varying based on the individual practitioner and the snow density. Samples in low density snow (<200 kg m^{-3}) are commonly biased high, while samples from high density snow (>400 kg m^{-3}) are commonly biased low (Proksch and others, 2016). In snow cores the uncertainties depend on the sample size, with an uncertainty of 5% for our typical 1000 cm^3 core section (Kelley and others, 1994)
- *Snow accumulation*: Site snow accumulation is calculated from snow depth and snow density, and is subject to sources of uncertainty therein. Combineduncertainty from these sources has been previously estimated to be 0.3 m w.e. (Beedle and others, 2014; Huss and others, 2009).
- *Snow and ice melt*: Site snow and ice melt is calculated from both the thickness of material lost and the density of that material, so the uncertainty is primarily due to surface roughness, snow density variability, stake deformation and drilling, and discrepancies between various types of observations (probe depths, stake length change, and snow pit depths). Combined uncertainty from these sources is estimated to be 0.10 m w.e. (Heinrichs and others, 1995).
- *Temperature*: Measurement uncertainty has decreased in time with sensor evolution. Through 1998, average daily temperatures were recorded using analog instruments with uncertainties estimated of ± 1 °C. After 1998, digital sensors with shorter response times and measurement intervals decreased measurement uncertainty to ± 0.25 °C (Kennedy and others, 1997).
- *Precipitation*: For the stage type gages (in use at Wolverine and Gulkana glaciers prior to 2016) measurement uncertainty is dominated by thermal expansion, with estimated daily uncertainty of 5–8 mm (Kennedy, 1995). The uncertainties are thought to be lower with weighing type Total Precipitation Gages (in use at Wolverine and Gulkana glaciers through 2022) and even lower with the radar disdrometers, but those are unquantified in these mountain environments. Published uncertainties for SNOTEL precipitation gauges range from 10-15% and are primarily attributed to under-catch, meaning that in windy environments accumulation type sensors often record less snow (Ikeda and others, 2021). Uncertainty estimates are not available for precipitation data acquired from the National Climate Data Center.

2. **Input_*Glacier*_Area_Altitude_Distribution.csv**

File containing time dependent glacier geometry provided as an Area Altitude Distribution (AAD) with the following columns:

- **Year:** Balance year in yyyy format. Calendar year in which the annual mass balance ends.
- **additional columns (*n*=11–23):** Columns 2 and higher have glacier area in square km within elevation bins. Column headers are the median elevation in meters for each bin (e.g., 250 m for a bin spanning 200–300 m elevation). Bin size is the difference between numeric column headers:100 m for Gulkana, Wolverine, and Lemon Creek Glaciers, 50 m for South Cascade Glacier,

and 30 m for Sperry Glacier. Each row in the data file provide the annual glacier area within each elevation bin in km^2. Glacier areas are derived from digital elevation models (DEMS; USGS data release https://doi.org/10.5066/P9R8BP3K) and area is interpolated linearly for years without DEM acquisitions (O'Neel and others, 2019). The total glacier area can be estimated by summing across the row for any given year.

3. **Input_*Glacier*_Geodetics.csv**
   File containing the acquisition date and properties of the DEMs used to construct AADs and calibrate mass changes for each glacier.
   - **Date:** Date of DEM acquisition in yyyy/MM/dd format. Date of the reference DEM (the DEM that other DEMs are differenced from) is identifiable by the 0.00 m w.e. entry in the **Mass_Change** column.
   - **Area_max:** The maximum glacier area during the geodetic mass balance time interval between the Date of DEM acquisition and the Date of the reference DEM in m^2.
   - **Area_min:** The minimum glacier area during the time interval between the Date of DEM acquisition and the Date of the reference DEM in m^2.
   - **dH:** The glacier-wide average elevation change during the time interval between the Date of DEM acquisition and the Date of the reference DEM in m. Note that this elevation change is computed over the Area_max.
   - **Mass_Change:** Geodetic mass balance for the time interval between the Date of DEM acquisition and the Date of the reference DEM in meters of water equivalent. Geodetic mass balance is computed by differencing digital elevation models to calculate glacier volume change and then converting volume to mass using the 850+/-60 kg m^{-3} material density assumption described by Huss (2013). Note that starting with version 8.0 the m w.e. geodetic mass change result uses the average glacier area over each geodetic interval. Details found in O'Neel and others (2019) and Florentine and others (2023).
   - **Uncertainty:** Uncertainty in m w.e. due to error associated with the elevation data (DEM) differencing required to compute glacier volume change, and the material density assumption required to convert ice volume change to water equivalent mass (Huss, 2013). Quantification of geodetic mass balance uncertainty is described fully in O'Neel and others (2019).

4. **Input_*Glacier*_Daily_Weather.csv**
   Tables in CSV format containing daily aggregated meteorological observations and station elevation data used in glacier-wide mass balance solutions. Includes data from stations operated by the USGS and from other continuously operated weather stations selected for proximity. Precipitation sensors vary between weather stations. The type of precipitation sensor is indicated in column headers if they are known, otherwise they are labeled 'Precip Other'. 'NaN' values indicate no data. Precipitation type columns are only included if there is data of that type associated with that glacier. Station summary, location information, and data sources are described further in Table 1. Note that historical weather data are subject to updates by host agencies as quality assurance techniques and methods evolve. Detailed information on USGS operated weather stations is available in an associated USGS data release https://doi.org/10.5066/P9EUXIPE.
   - **Date:** observation date in yyyy/MM/dd format
   - **Temperature:** Daily average temperature in degrees C. At USGS sites, average calculated as arithmetic mean daily temperature. At non-USGS sites, the average is calculated using the World Meteorological Organization (WMO) method of dividing the sum of the daily minimum and maximum values by two.
   - **Precip_Stage:** Daily total measured precipitation in millimeters of water equivalent (mm w.e.) for USGS stage-type instrument.
   - **Precip_TPG:** Daily total measured precipitation in millimeters of water equivalent (mm w.e.) for USGS operated, liquid filled, load cell style total precipitation gauge.

- **Precip_Other**: Daily total measured precipitation in millimeters of water equivalent (mm w.e.) for non-USGS stations, utilizing unknown instrument types.
- **Precip_Disdrometer:** Daily total measured precipitation in millimeters of water equivalent (mm w.e.) for USGS operated radar disdrometer sensors.
- **Elevation**: Approximate weather station elevation in meters above sea level.

Glacier	Approximate Elevation (m)	Site Name	Source	Latitude	Longitude
Gulkana	481	Gulkana Airport	NCDC	62.1592	-145.4589
Gulkana	1480	Moraine	USGS	63.2614	-145.4102
Gulkana	1725	Nunatak	USGS	63.2818	-145.4260
Wolverine	4	Seward Airport	NCDC	60.1292	-149.4186
Wolverine	990	Wx990	USGS	60.3819	-148.9397
Lemon Creek/Taku	5	Juneau Airport	NCDC	58.3566	-134.5640
South Cascade	270	Diablo Dam	NCDC	48.7141	-121.1430
Sperry	963	Hungry Horse Dam	NCDC	48.3425	-114.0217
Sperry	1920	Flattop SNOTEL	NRCS	48.8017	-113.8570

Table 1: Summary of weather station information for stations used for each glacier within the data release. USGS operates weather stations at Gulkana and Wolverine Glaciers that are within the glacier basin; others are continuously operating weather stations selected for proximity to the glacier. Coordinates are given in decimal degrees; elevations are in meters. Source is the organization that operates each station. NCDC = National Climatic Data Center (see Menne and others, 2013); NRCS = National Resources Conservation Service (see USDA, 2026); USGS = U.S. Geological Survey

To calculate glacier-wide mass balance, we use a temperature-precipitation index model, as described in O'Neel and others (2019). We use the model to estimate the dates of seasonal maximums and minimums, as well as the mass changes between those dates and the direct measurements.

Before Version 9.0: A single, primary weather station record was selected for each glacier based on duration, continuity, and proximity. Data from secondary (more distal) stations filled primary weather station data gaps via a linear regression by month between the primary and secondary data (van Beausekom and others, 2010). Reliance on these "nested regression" to fill data gaps increased the uncertainties of the mass balance. Increasingly large portions of the daily weather record published in Version 8 of this data release were based on regressions with distal weather stations rather data from the primary weather station as glacier recession forced the abandonment of the original Gulkana weather station.

Beginning with Version 9.0 of this data release, we avoid the nested regressions and attendant uncertainties by separating temperature and precipitation measurements from individual stations and sensor types (as listed in Table 1). Ablation and precipitation coefficients are calculated directly for each weather station. For each time-period of interest (e.g., the interval between a glaciological measurement and the glacier-wide seasonal maximum or minimum), weather data with the highest predictive capability—typically from the closest observations—are used to estimate the mass changes. All other aspects of the daily weather routines remain consistent with O'Neel and others (2019). Short gaps in daily temperature data (less than 3 days) are filled via linear interpolation based on temperatures recorded immediately before and after the gap. In cases where the model needs to extrapolate beyond the available data (such as for future projections), mean values for that day-of-year are used. This is mainly relevant when the model runs ahead of provisional data updates for the current mass-balance year. Temperature lapse rates are assumed to be -6.5°C/km. However, since Version 5.0, lapse rates for Lemon Creek Glacier

has been optimized to -5.0°C/km based on data from local weather stations, as described in McNeil and others (2020).

5. **Input_*Glacier*_SubSeasonal_Glaciological_Data.csv**
   Tables in CSV format containing containing sub-seasonal glaciological measurements, if sub-seasonal measurements have been collected at that glacier. Where available, they are used to better constrain the mass balance calibration. Further details are available in supplemental materials of O'Neel and others (2019) and McNeil and others (2020).
   - **Year:** Balance year in yyyy format. Calendar year in which the annual mass balance ends.
   - **site_name:** Unique identifier of each mass balance measurement site on each glacier. Sites have specific geographic coordinates that are resampled each year, and correspond to the "site_name" variable in other data files included in this data package. Maps of all sites are provided in the README and site coordinates are provided in the *Glacier_Mass_Balance_Data_Sites.zip* included in this data package.
   - **Elevation:** Elevation in meters above ellipsoid, taken from a digital elevation model (DEM) in the year of observation, or otherwise interpolated between DEM dates (O'Neel and others, 2019).
   - **Date1:** Date of first observation in YYYY/MM/DD format.
   - **Surface1:** Material on surface at time of Date1 observation.
   - **Date2:** Date of second observation in YYYY/MM/DD format.
   - **Surface2:** Material on surface at time of Date2 observation.
   - **db:** Change in mass balance between Date1 and Date2, in m w.e. (O'Neel and others, 2019).

6. **Input_*Glacier*_TSL_Date.csv**
   Table with transient snow line elevation time-series data for Lemon Creek and Taku Glaciers, necessary due to the mid-summer timing of glaciological measurements prior to 2016. Snow line location is determined visually on the centerline of the glacier in satellite imagery, and the elevation is extracted from a DEM. Snowlines are primarily from Landsat, with the addition of some Sentinel and Worldview images in more recent years. Further details are available in supplemental materials of McNeil and others (2020).
   - **Year:** Balance year in YYYY format
   - **Date:** Date of the satellite image in MM/DD/YYYY format
   - **Elevation:** Elevation in meters given as height above ellipsoid.

OUTPUT DATA

1. **Output_*Glacier*_Glacier_Wide_solutions_calibrated.csv**
   CSV tables of annual and seasonal (winter and summer) glacier-wide mass balance solutions calibrated with geodetic observations (O'Neel and others, 2019; Florentine and others., 2023). 'NaN' values indicate no data.

   - **Year**: Balance year in YYYY format. Calendar year in which the annual mass balance ends.
   - **Bw:** Glacier-wide winter mass balance, in m w.e.
   - **Bs:** Glacier-wide summer mass balance, in m w.e.
   - **Ba:** Glacier-wide average annual mass balance, in m w. e.
   - **ELA:** Equilibrium line altitude (i.e., elevation where the mas balance profile is equal to 0), as modeled during the calculation of glacier-wide mass balance, in meters above sea level.
   - **Bw_Date**: Date of mass maximum. Modeled estimate for the day of mass maximum, using local precipitation, temperature data, and a degree day model.
   - **Ba_Date**: Date of mass minimum. Modeled estimate for the day of the mass minimum, using local precipitation, temperature data, and a degree day model.
   - **Calibration:** geodetic calibration, given in units of m w.e.

Mass Balance Model

Weather data are used to model the timing of mass extrema and calculate mass changes between measurement dates and the modeled mass extrema for each site on the glacier using a standard temperature-index melt model (or 'degree day' model; e.g., Hock, 2003). Prior to Version 9, mass balance modeling was performed using a single weather record, where data gaps were infilled by a secondary distal weather station. A single set of degree-day factors and precipitation ratios was then applied to this weather data to model on-glacier estimates of mass balance. For Version 9 forward, weather stations and their respective temperature and precipitation sensors are independently calibrated to in-situ measurements. Then, for each instance where the mass balance model is applied, daily weather station data are selected based on proximity and predictive capacity. The selection criteria include the proximity of each glacier site to a weather station based on elevation, and the R-squared value obtained when calibrating degree-day factors and precipitation ratios. Weather data selection occurs at daily time-steps, where the respective degree-day factors and precipitation ratios are applied to the best available temperature and precipitation data. Finally, a cumulative mass-balance time-series is derived by summing daily values to provide the respective mass change between measurement and mass extrema dates.

Mass Balance Profile

Glacier-wide mass balance results are produced using a piece-wise linear balance profile describing the relationship between mass balance and elevation, with a single break-point typically near the long-term ELA. This piece-wise balance profile was chosen because it incorporates physical expectations of variable melt rates for snow and ice. Our choice was guided by several factors including: 1) feasibility and goodness-of-fit at all glaciers, 2) applicability for both sparse and robust data sets, and 3) capacity to include short-lived expanded stake and ground-penetrating radar campaigns above long-term index sites. Each fit is constrained to have a single interior breakpoint, or knot, in the piecewise-linear fit. Further details are given in O'Neel and others, 2019.

Geodetic Calibration

USGS follows much of the re-analysis procedure outlined by Zemp and others (2013). For geodetic calibration, however, we adopt a piecewise/ breakpoint fitting approach over approximately decadal intervals. This maximizes the incorporation of geodetic data, while minimizing the impacts of non-stationarity that manifest in multi-decadal calibrations and limits the impact of errors arising from material density assumptions. Note that starting with version 8.0 this m w.e. geodetic mass change result reflects time-varying glacier area, i.e., the glacier area assignment is not fixed but uses the average glacier area over the geodetic interval. For further details, see O'Neel and others (2019) and Florentine and others (2023). Associated geodetic data used to derive the calibrations are

given in an accompanying data release, found under the USGS Benchmark Glacier Mass Balance and Project Data collection via https://doi.org/10.5066/F7BG2N8R.

SUGGESTED CITATION:

Where possible, please cite larger mass balance project data collection, of which this release is a part:

U.S. Geological Survey Glacier Project, 2020, USGS glacier project comprehensive data collection: U.S. Geological Survey data release, https://doi.org/10.5066/P9AGXQSR

This dataset can be cited separately, if needed, as:

U.S. Geological Survey Glacier Project, 2016, Compiled Input Data and Glacier-Wide Mass Balances (ver. 10.0, March 2026): U.S. Geological Survey data release, https://doi.org/10.5066/F7HD7SRF

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